Intro to JS

Define/explain the following(based on the lesson and/or research from the internet):

1. Variables Used to store data that can change.
2. Constants Used to store data that cannot change.
3. Data Types

   Number: numbers

   String: text

   Boolean: true or false

4. Numbers Examples: 1, 3.14, -10
5. Strings Text in quotes: "Hello" or 'World'
6. Booleans true or false
7. Let statements Declare a variable and assign a value.
8. Variable Naming

   Can contain letters, numbers, underscores, $

   Cannot start with a number

   Often uses camelCase: myVariable

   Math Operators
9. * multiply
10. ** exponent
11. Math.PI constant π
12. % modulus
13. x=x+1 increment
14. X++ increment by 1
15. a*=2 multiply and assign
16. a - - decrement by 1
17. / divide

What are the final values of all variables a, b, c and d after the code below?

```
let a = 1

let b = 1;

b++;

c=b;

d=a+b;
```

A **Trace Table** helps you keep track of each variable's value after execution of each line of code. Match your predicted results with the following trace table (the last line always shows the final

values of all variables):

🔍 Trace Table:

Step	Statement	a	b	c	d
1	`let a = 1`	1			
2	`let b = 1`	1	1		
3	`b++`	1	2		
4	`c = b`	1	2	2	
5	`d = a + b`	1	2	2	3

What are the values of a and x after the code below?

let a = 2;

let x = 1 + a * 2;

Here's its Trace Table (compare to your results)

🔍 Trace Table:

Step	Statement	a	x
1	`let a = 2`	2	
2	`let x = 1 + a * 2`	2	5

What are the final values of all variables x, y, z and w after the code below?

```
let x = 5;

let y = x + 2;

y--;

z = y;

w = x * y;
```

Fill in the following Trace Table for the last program:

Step	Statement	w	x	y	z

Determine the final values of `m`, `n`, `o`, and `p` after the following JavaScript code executes:

```
let m = 10;

let n = m - 3;

n += 2;

o = n;

p = m + o;
```

Fill in the following Trace Table for the last program:

Step	Statement	m	n	o	p

After the following code runs, what are the values of `i`, `j`, `k`, and `l`?

```
let i = 0;

let j = 4;

j = j / 2;

k = j;

l = i + j + k;
```

Fill in the following Trace Table for the last program:

Step	Statement	i	j	k	l

What will be the values of `r`, `s`, `t`, and `u` at the end of this script?

```
let r = 7;

let s = r;

s *= 2;

t = s - r;

u = r + s + t;
```

Find the final values of `a`, `b`, `c`, and `d` after this code executes:

```
let a = 3;

let b = a + 1;
```

```
b--;

c = a * b;

d = c - a;
```

Complete the following programming exercises. Include a screenshot of your solution following each problem. Some of the questions will have partial code. Fill in the missing pieces to make it work.

Exercise :Volume of a Cylinder

- Create a program that displays the volume of a cylinder with a height of 10cm and a diameter of 15cm.

```
let volume=0;
let height=10;
let diameter=15;
let radius=_____;

volume=Math.PI*radius**2*height;

console.log("The volume of  a cylinder with a height of " + height + " cm and diameter of " + diameter + " cm is " + volume + " cm cubed.");
```

Exercise : Surface Area of a Cube

- Create a program that displays the surface area of a cube with a side length of 2.25m.

```
let side=2.25;

let areaOfOneSide=side*side;

let surfaceArea=side*6;
```

```
console.log("The surface area of a cube with a side length of " + _______+ " cm is " +
_____________ + " cm squared.");
```

Exercise : Investment

- Create a program that determines how much money you would have in an investment if you invested $10,986.98 for 5 years at 4.3%. The formula for this is $A=P(1+i)^n$, where P in your principal investment, i is the interest rate as a decimal and n is the number of years.

```
let principal=10986.98;

let term=5;

let rate=4.3;

let amount=_____*(1+_____/100)**_____;

console.log("If you invest $"+ principal +" for "+term + " years at " + rate +"%, it
will grow to $"+amount);
```

Exercise : Hours to Seconds

Write a program that calculates the number of seconds in 4.5 hours.

```
_____ secondsInHours=60*60*60;

let hours=_____;

let totalSeconds=_____*_____;

console.log("The total number of seconds in "+hours+" hours is "+totalSeconds);
```

Exercise : Area of a Triangle

Write a program that displays the area of the triangle with a base length of 10cm and a height of 15cm.

Exercise : Area of a Circle w/ Radius

Write a program that displays the area of a circle given the radius of 24.98 mm.

Use the constant `Math.PI` from the `Math` class.

Exercise : Square a Double Type

Write a program that computes the square of the number 1221.

Exercise : Cube of a Number

Write a program that computes the cube of the number 1221.

Exercise : Weight of a Car Carrier Trailer

Write a program that displays how much the weight is on a car carrier trailer given the average weight of a car is 1220 kg and number of cars on the trailer is 101.

Exercise : Purchase Calculator

Create a simple calculator that calculates the pre-tax amount of purchased items at a store that cost $4.99, $7.99, $3.99 and $11.95.

Exercise : Product

Write a program that displays the product of 102 and 2155.

Exercise : Cubed

Write a program that displays the cube of 42.

Exercise : Average

Write a program that displays the average of 11,22,33 and 44.

Exercise : Circumference of Circle

Write a program that displays the circumference of a circle given the radius of 98.7 m! Assume pi is 3.14.

Note: the equation for circumference is `2pi*radius`.

Exercise : Volume of Sphere

Write a program that displays the volume of a sphere given the radius of 1100 cm.
Assume pi is 3.14.

Note: The equation for a sphere's volume is (4/3)*pi * radius**3

Exercise : Pythagorean

Write a program that displays the length of the hypotenuse by using the Pythagorean theorem assuming the lengths of sides a and b are 10 cm and 12 cm..

The Pythagorean Theorem is $a^2 + b^2 = c^2$.